

Learning Intention: I can describe the properties of solids and liquids and explain how heat energy causes a change of state.

Success Criteria — I can:

- Identify observable properties of solids and liquids.
- Explain what happens when heat energy is added or removed.
- Give everyday examples of a change of state.

Reading Strategy

Before reading, look at the heading and any bold words — they tell you what the text is about. As you read, look for **facts** (what something is) and **cause-and-effect** (what happens when something changes).

Read — Solids, Liquids and Changing State

Everything around us is made of matter, and matter can exist in different states. Two states you see every day are **solid** and **liquid**.

A **solid** has a fixed shape. It doesn't flow, and it keeps its shape whether it's inside a container or sitting on a table — think of a block of ice, a wooden block, or a rock. A **liquid** does not have a fixed shape. It flows and takes the shape of whatever container it's in, like water poured into a cup or juice poured into a bottle. However, a liquid does keep the same amount, or volume, even as its shape changes.

Matter can change from one state to another when heat energy is added or removed. When you **add** heat energy to a solid, like ice, it can **melt** and become a liquid, like water. When you **remove** heat energy from a liquid, like water, it can **freeze** and become a solid, like ice.

This happens all around us. Chocolate melts in warm hands. Butter melts in a hot pan. Icy poles melt on a hot day. In the freezer, liquid water freezes into solid ice cubes. Each of these is a **change of state** caused by adding or removing heat.

Vocabulary Helper

Word	Meaning
Melt	Change from a solid to a liquid by adding heat
Freeze	Change from a liquid to a solid by removing heat
Change of state	When matter changes from one form to another (e.g. solid to liquid)

Parent Tip

Watch an ice cube melt together at home — talk about what's happening to the water molecules as heat energy is added. You could also freeze a cup of water in the freezer and observe the reverse change.

Did You Know?

Not all solids melt at the same temperature! Ice melts at 0°C, but metals like iron need to be heated to over 1,500°C before they melt. Chocolate melts at a much lower temperature than both — that's why it can melt just from being held in a warm hand.

Find It, Think About It & Apply

Q1 — Find It (Literal)

- a) What happens to a solid when heat energy is added? _____
- b) What happens to a liquid when heat energy is removed? _____
- c) Name three everyday examples of melting mentioned in the text.

Q2 — Think About It (Inferred)

- a) Why do you think chocolate melts so easily in a warm hand? _____
- b) Why does a liquid change shape when poured into a different container, but a solid does not?

Q3 — Sort the Properties

Write whether each description matches a solid or a liquid.

Description	Solid / Liquid
Has a fixed shape that doesn't change	
Flows and takes the shape of its container	
Keeps the same volume, but not the same shape	

Q4 — Match Cause to Effect

Cause	Effect (write the letter)
Heat added to ice	
Heat removed from liquid water	
Heat added to a chocolate bar	

A = Freezes into solid ice B = Melts into a puddle C = Melts into liquid water

Watch Out!

Melting and freezing are opposite changes, but they involve the **same** substance changing state — water can be liquid or solid (ice), but it's still water. Adding heat energy doesn't create a new substance.

Q5 — True or False?

Statement	True / False
When ice melts into water, it becomes a completely different substance.	True / False
A liquid keeps the same volume even when it changes shape.	True / False

Apply & Reasoning — Use what you've read to explain your thinking.

- 1 Reasoning:** Compare a solid and a liquid. Give one way they are different and explain why this difference matters in everyday life.

- 2 Reasoning:** In your own words, explain what causes a solid to change into a liquid.

- 3 Open-ended:** Think of a solid you know that can melt (or a liquid that can freeze). Describe its properties as a solid, then describe how it changes when heat is added or removed.

Quick Check — Exit Ticket

Name one property that helps you tell a solid apart from a liquid.

Answer & Teacher Support

Answer Key — Pages 1 & 2

Q1 (p2): a) It melts and becomes a liquid. b) It freezes and becomes a solid. c) Chocolate, butter and icy poles.

Q2 (p2): a) Chocolate has a low melting point, so the warmth of a hand alone is enough heat energy to melt it. b) A liquid flows and has no fixed shape of its own, so it takes the shape of its container; a solid holds its own shape.

Q3 (p2): Fixed shape — Solid. Flows, takes shape of container — Liquid. Same volume, changing shape — Liquid.

Q4 (p2): Heat added to ice — C. Heat removed from water — A. Heat added to chocolate — B.

Q5 (p2): "Ice melting becomes a different substance" — False (still water). "Liquid keeps same volume, changes shape" — True.

Answer Key — Page 3

Q1 (acceptable-answer guide): Accept any valid difference (e.g. solid keeps its shape, liquid flows) linked to a sensible everyday example (e.g. why we can stack solid blocks but not liquid).

Q2: Adding heat energy to a solid causes it to melt and become a liquid.

Q3 (acceptable-answer guide): Accept any response describing plausible solid properties and a correct, clear explanation of the melting or freezing change involved.

Exit ticket: Accept any valid property, e.g. a solid keeps its shape while a liquid flows and takes the shape of its container.

Teacher Checklist

- ☐ Retrieves literal facts from the nonfiction text.
- ☐ Makes reasonable inferences beyond the text.
- ☐ Sorts solid and liquid properties accurately.
- ☐ Links heat energy added/removed to the correct change of state.
- ☐ Evaluates true/false statements about states of matter.
- ☐ Explains a change of state in their own words with an example.

Teacher Tip

This resource combines nonfiction reading comprehension with AC9S3U04 (solids, liquids and change of state) — a genuine curriculum gap this fills. Pair with a real ice-melting or butter-melting observation if hands-on materials are available.